

How to determine the velocity resolution of a radio telescope spectrometer?

I came across an exercise to confirm the velocity resolution of SEQUOIA receiver of TRAO telescope, which has frequency resolution of 15 kHz. To convert frequency resolution to velocity resolution, we start from the high school light doppler effect formula for stationary observer (Cutnell, 2011, Equation 16.12):

$$f_o = f_s \left(\frac{1}{1 + \frac{v_s}{c}} \right)$$

We want to find that is the velocity v_s to get a frequency shift of 15 kHz, so

$$\Delta f = f_o - f_s = 15 \text{ kHz}$$

$$\Delta f = f_o - f_s = f_s \left(\frac{1}{1 + \frac{v_s}{c}} \right) - f_s = f_s \left(\frac{1}{1 + \frac{v_s}{c}} - 1 \right)$$

Rearrange the equation and put v_s as subject to get:

$$v_s = \frac{c}{\left(\frac{\Delta f}{f_s} + 1 \right)} - c$$

Which can be further simplified into:

$$v_s = - \frac{c \Delta f}{\Delta f + f_s}$$

Similarly:

$$\Delta f = \frac{v_s f_s}{-c - v_s}$$

Sometimes seen as $v_s = c \frac{\Delta f}{f_s}$ on internet because only concern about magnitude and Δf is very small compared to f_s so $\Delta f + f_s \approx f_s$.

So, the velocity resolution of TRAO SEQUOIA at C18O (J=1-0) is -41.32m/s $\approx$ -0.05 km/s, the negative means the inverse relationship between source velocity and observed frequency.

Reference:

Cutnell, J. D. (2011). *Introduction to Physics 8th Edition International Student Version with WP V5 Set*. Wiley.